

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

Register Number	192511009
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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 10%

QUESTIONS:2

Total Marks:20

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I _Avishek N (192511009) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Avishek N

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

ANSWER :

An intelligent agent perceives its environment through sensors and acts upon it through actuators. In the smart home automation system described, the agent must handle lighting, temperature, and security by reacting to motion, temperature changes, time of day, and learned user preferences. The four classical agent types can be examined in this context as follows.

1. Simple Reflex Agent

A simple reflex agent selects actions based only on the current percept, using a fixed set of condition-action (if-then) rules, with no memory of past states.

Working in the system: If motion is detected → turn on the light. If temperature > 28°C → turn on the AC. If door is opened while armed → trigger alarm.

Limitation: It cannot learn user habits or distinguish between an authorized person and an intruder, and it may act wastefully (e.g., turning on lights in an empty room briefly triggered by a pet).

2. Model-Based Reflex Agent

This agent maintains an internal model of the world to track aspects of the environment that are not directly observable at the current instant, such as whether a room is currently occupied or whether it is day or night.

Working in the system: It keeps track of "room occupied" status even between motion triggers, and combines this with the time of day to decide whether to dim, switch off, or keep lights on.

Advantage over simple reflex: Reduces false actions and unnecessary switching by considering recent history/state, not just the instantaneous percept.

3. Goal-Based Agent

A goal-based agent chooses actions that help it achieve a specific defined goal, using search or planning rather than fixed rules.

Working in the system: Given the goal "maintain security while occupants are away," the agent plans a sequence of actions - lock all doors, arm sensors, switch off unnecessary appliances - rather than reacting to one event at a time.

Advantage: It can handle complex, multi-step objectives such as "prepare the house for the night" that a reflex agent cannot express.

4. Utility-Based Agent

A utility-based agent goes further by assigning a utility value to each possible state and choosing the action sequence that maximizes overall satisfaction, allowing it to balance competing objectives.

Working in the system: It balances comfort, energy efficiency, and security together - for example, choosing a temperature setting that keeps the room comfortable while minimizing power consumption, or deciding how bright to keep security lighting without wasting energy. Advantage: It can make trade-offs between conflicting goals (comfort vs. energy saving) rather than treating every goal as equally binding.

Comparative Analysis

Agent Type	Memory of Past	Handles Goals	Handles Trade-offs	Suitability Here
Simple Reflex	No	No	No	Low – too rigid, reacts poorly to changing context
Model-Based Reflex	Yes (internal state)	No	No	Medium – good for tracking occupancy/time
Goal-Based	Yes	Yes	No	Medium-High – good for security routines
Utility-Based	Yes	Yes	Yes	High – best for balancing comfort, energy, security

Justification / Recommended Design

A single pure agent type is not ideal for this system. The most effective design is a hybrid model-based + utility-based agent:

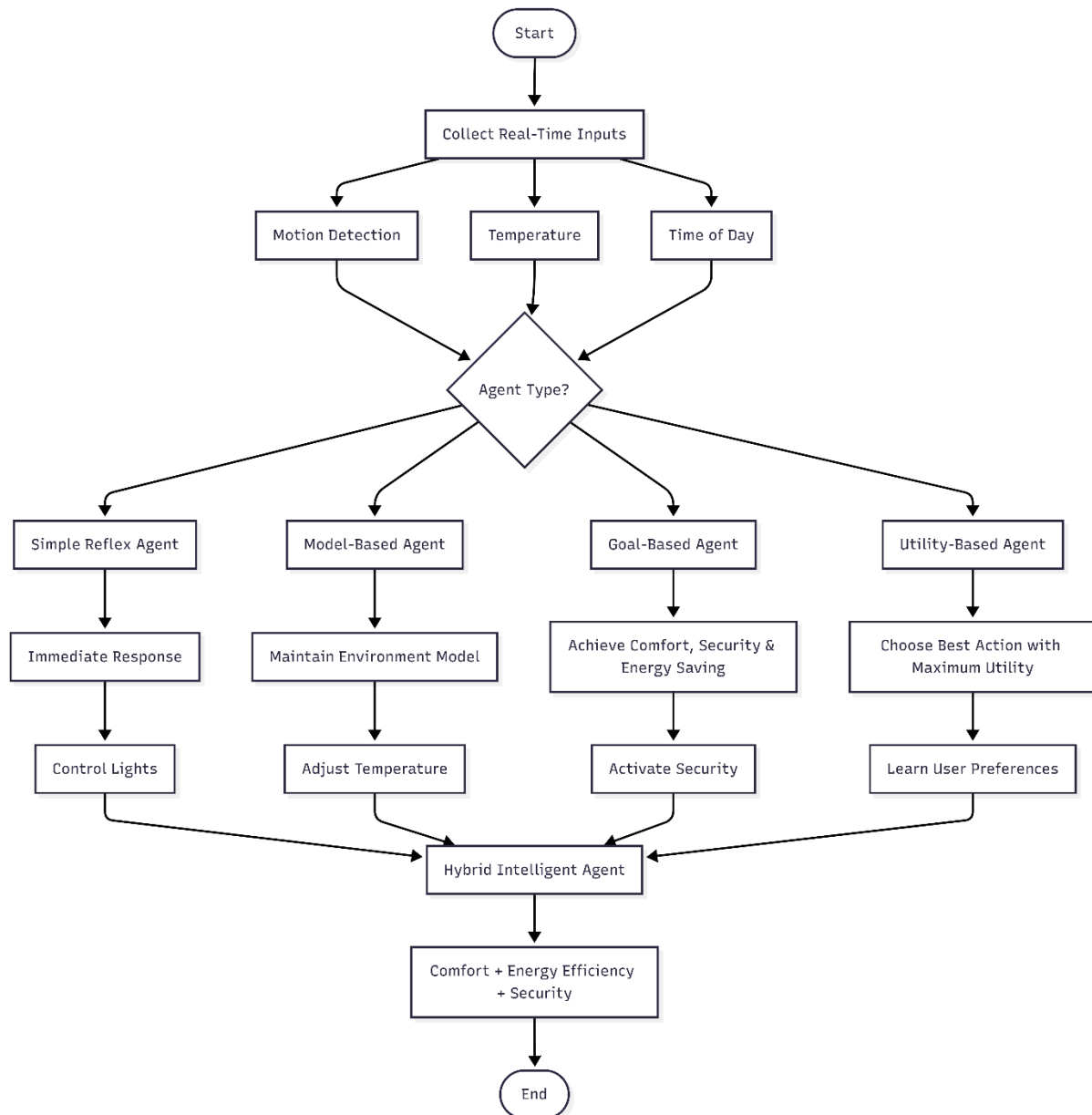
The model-based component continuously maintains an internal state - room occupancy, time of day, current temperature trend, and learned user routine - so decisions are context-aware rather than purely reactive.

The utility-based component uses this internal state to score alternative actions (e.g., "AC at 24°C now" vs. "AC at 26°C, delayed by 10 minutes") and picks the one that best balances comfort, energy efficiency, and security, and can also incorporate learning over time to refine what the user prefers.

This hybrid approach gives the responsiveness of a reflex system, the situational awareness of a model-based agent, and the ability to reconcile competing priorities that only a utility-based agent provides - making it the most suitable choice for a comfort-, energy-, and security-aware smart home.

Advantages:

- Improves user comfort by automating home appliances.
- Saves energy by reducing unnecessary power consumption.
- Enhances home security through continuous monitoring.
- Learns user preferences for better decision-making.
- Reduces manual effort and increases convenience.



Disadvantages

High installation and maintenance cost.
Depends on sensors and internet connectivity.
Privacy and security risks due to data collection.
System failure may affect home operations.
Requires regular software updates.

Conclusion

A hybrid intelligent agent is the best choice for a smart home automation system because it combines the strengths of simple reflex, model-based, goal-based, and utility-based agents. It provides better comfort, improves energy efficiency, enhances security, and continuously learns user preferences for smarter decision-making.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

ANSWER

The robot exploring an unknown maze has no map and no heuristic information about which direction leads closer to the exit, so only uninformed (blind) search strategies are applicable. Two suitable strategies are Uniform Cost Search (UCS) and Iterative Deepening Search (IDS).

Uniform Cost Search (UCS)

UCS explores the maze by always expanding the node with the lowest cumulative path cost $g(n)$ so far, using a priority queue (min-heap) ordered by path cost rather than a simple FIFO queue. This guarantees the cheapest path to the exit is found, even if move costs (e.g., diagonal vs. straight moves) differ across the maze.

Iterative Deepening Search (IDS)

IDS repeatedly performs a Depth-Limited Search (DLS), starting with depth limit 0 and increasing the limit by 1 each time the exit is not found, until the exit is located. It combines the low memory usage of Depth-First Search with the completeness/optimalty (for uniform step cost) of Breadth-First Search.

Time and Space Complexity

Algorithm	Time Complexity	Space Complexity	Complete?	Optimal?
Uniform Cost Search (UCS)	$O(b^{(1 + \lceil C^*/\epsilon \rceil)})$	$O(b^{(1 + \lceil C^*/\epsilon \rceil)})$	Yes (finite b , positive step cost)	Yes (least-cost path)
Iterative Deepening Search (IDS)	$O(b^d)$	$O(b \times d)$	Yes (finite b)	Yes, if step cost is uniform

Here b = branching factor (number of open cell directions per node), d = depth of the shallowest exit, C^* = cost of the optimal path, and ϵ = the smallest step cost in the maze.

Advantages and Disadvantages:

UCS advantage: Always finds the cheapest path even when step costs vary (e.g., rough terrain in parts of the maze); UCS disadvantage: needs more memory to maintain the priority queue and can explore many nodes if costs are similar (large effective branching).

IDS advantage: Uses only linear space $O(b \times d)$, making it far more memory-efficient for large mazes; IDS disadvantage: repeats work at every depth iteration, and it is only optimal if every move has equal (uniform) cost - if costs differ, UCS is preferred.

Relating the Problem to Intelligent Agent Types

The maze-solving robot is best modelled as a goal-based agent, since its objective ("reach the exit") is explicit and requires planning a sequence of moves rather than reacting to single percepts. A simple reflex agent (e.g., "if wall ahead, turn right") could get trapped in loops in a complex maze, and a model-based reflex agent, while tracking visited cells, still lacks an explicit search process toward a goal. A utility-based agent becomes relevant if the robot must also minimize battery usage or time taken, since it would then need to weigh path cost against resource consumption, not just reachability - agent design directly affects decision-making by determining whether the robot merely reacts, tracks state, pursues an explicit goal, or optimizes among several competing objectives.

Recommended Combination and Justification

The most effective combination is a goal-based agent using Iterative Deepening Search as the default strategy, since the maze is unknown, memory is limited on a physical robot, and the robot has no prior knowledge of maze size or depth to the exit - IDS's linear space usage and guaranteed completeness make it well suited to this uncertainty.

However, if different maze cells have varying traversal costs (e.g., slippery floor, rough patches, or narrow gaps costing more time/energy), a goal-based agent using Uniform Cost Search should be used instead, since only UCS guarantees the truly cheapest path under non-uniform costs. In practice, a goal-based agent that starts with IDS to explore quickly and switches to UCS logic once cost information about the maze is known offers the best balance of memory efficiency and path optimality.

